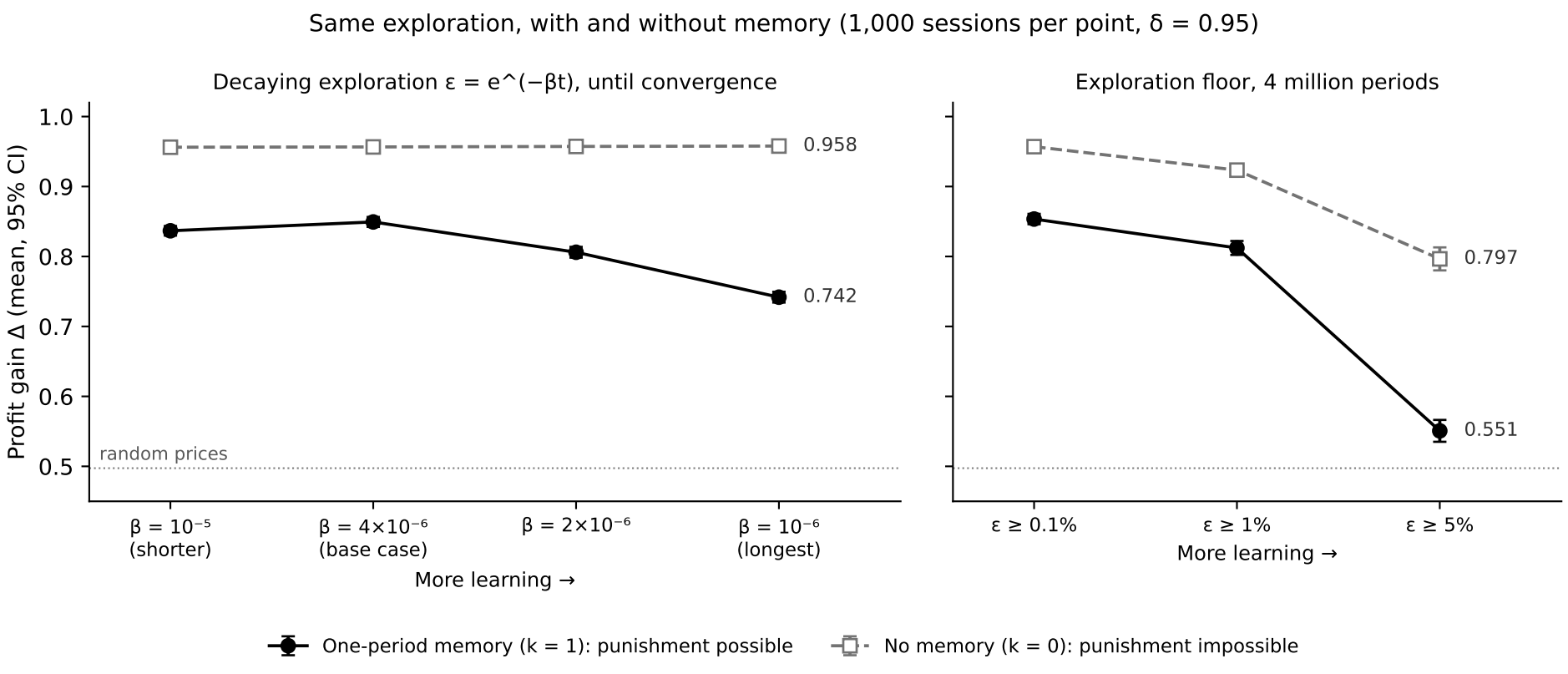
Same exploration, with and without memory (1,000 sessions per point, δ = 0.95)
Decaying exploration ε = e^(−βt), until convergence
Exploration floor, 4 million periods
Profit gain Δ (mean, 95% CI)
1.0
0.9
0.8
0.7
0.6
0.5
0.958
0.742
0.797
0.551
random prices
β = 10⁻⁵ (shorter)
β = 4×10⁻⁶ (base case)
β = 2×10⁻⁶
β = 10⁻⁶ (longest)
More learning →
ε ≥ 0.1%
ε ≥ 1%
ε ≥ 5%
More learning →
One-period memory (k = 1): punishment possible
No memory (k = 0): punishment impossible